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Master Business Specification & Technical Architecture

Next-Generation E-Commerce Auto-Lubricant Platform with Multi-Agent AI

Algerian Automotive Retail Project

September 11, 2026

Abstract

This document details the business logic, architectural blueprint, database schema, and multi-agent AI framework for a hybrid online/offline e-commerce platform specializing in automotive lubricants and fluids within the Algerian market. By embedding an interactive "Guided Selling" AI technician directly into the customer journey, the platform bridge the technical knowledge gap for vehicle maintenance while offering standard dual-path navigation, localized vehicle mapping, dynamic self-learning specification databases, and low-friction order placement.

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1 Executive Summary & Market Context

The primary objective of this project is to construct an e-commerce platform linked to a physical automotive lubricant retail store operating in Algeria. Selecting correct vehicle fluids—such as engine oil, gearbox (transmission) fluid, brake fluid, and coolant—requires exact alignment with vehicle manufacturer specifications (OEM standards), engine displacement, viscosity, and precise sump volume capacity.

1.1 Target Market Dynamics (Algeria)

- **High-Volume Vehicle Fleet:** Dominance of specific European and Asian vehicle models (e.g., Renault Symbol, Dacia Logan, Peugeot 208/301, Hyundai Accent, Volkswagen Golf, Toyota Hilux).
- **Friction in Offline Purchase:** Customers frequently buy incorrect oil viscosity or insufficient volume due to confusing engine codes and lack of technical manuals.
- **Commercial Flow Preference:** Low adoption of online credit card processing. Preferred checkout consists of simple order registration (Name, Phone, Wilaya, Address) followed by direct phone confirmation by the shop management prior to local delivery or Cash-on-Delivery (COD).

1.2 Core Differentiators

1. **Dual-Path Discovery:** Traditional search bar and multi-attribute catalog filtering for mechanics/experts alongside an interactive Multi-Agent AI Assistant for everyday drivers.
2. **Automated Volume Round-Up Math:** Precise calculations mapping required engine capacities (e.g., 4.3 L) directly to available container sizes (e.g., 1×5 L bottle or 1×4 L + 1×1 L combination).
3. **Self-Learning Database Architecture:** Automated fallback web search for missing vehicle models that populates a staging database for manual administrative verification.

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2 System User Experience & Order Workflow

The platform supports two distinct navigation paths that converge into a unified shopping bag and low-friction checkout workflow.

2.1 Dual Discovery Paths

- **Direct Catalog Path:** Experienced buyers search directly by keyword ("*Total Quartz 9000 5W-40*"), brand filters (*TotalEnergies, Liqui Moly, Naftal, Castrol, Mobil*), viscosity, or OEM approval standards (*RN0700, VW 507.00, PSA B71 2290*).
- **Guided AI Assistant Path:** Non-expert users engage with an AI chat assistant. The agent asks multi-turn follow-up questions to identify the vehicle's Make, Model, Year, and Engine Reference, automatically fetches matching inventory items, and renders interactive product cards.

2.2 Product Cards & Shopping Bag Interaction

1. Clicking an interactive card generated inside the AI chat stream routes the user to the Catalog Page pre-filtered for that precise fluid match.
2. Clicking "Add to Bag" increments the item count in the Shopping Bag drawer without triggering modal interruptions.
3. Opening the Shopping Bag drawer displays total selected volume, itemized costs, and the "Order Now" button.
4. Clicking "Order Now" opens a minimal submission form capturing:

$$\text{Order Payload} = \{\text{Full Name, Phone Number, Wilaya/City, Delivery Address}\} \quad (1)$$

5. Submitting the form creates an entry with status **Pending Confirmation** in the store backend. Staff review the order specs and perform confirmation via telephone.

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3 Multi-Agent AI System Architecture

The conversational engine employs a multi-agent framework built with specialized system prompts, toolsets, and safety evaluation guardrails.

3.1 Agent Roles & Responsibilities

- **Router Agent:** Conducts conversational slot-filling. Interrogates the user until Make, Model, Year, Engine Code, and desired Fluid Type (Engine, Gearbox, Brake, Coolant) are fully extracted. Outputs clean JSON data.
- **Database Search Agent:** Translates structured vehicle JSON into SQL queries. Queries the local `cars` table to retrieve viscosity and OEM specs, then queries the `inventory` table to find available stock. Performs container volume calculations.
- **Web Search Agent:** Triggered strictly when a vehicle model or engine code does not exist in the local database. Scrapes technical sources, extracts specifications, and writes to a staging table (`is_verified = FALSE`).
- **Evaluator & Handoff Agent:** Monitors requests involving vehicle state changes (e.g., high mileage > 200,000 km, heavy oil consumption, or mechanic-recommended viscosity changes). Scores recommendation confidence:
 - **Confidence** $\geq$ 80%: Displays safe alternative product cards with technical disclaimers.
 - **Confidence** < 80%: Halts automated suggestions and displays human store support contact details (Phone / WhatsApp).

3.2 Multi-Agent Orchestration Flow

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4 Database Schema Design

The platform relies on a normalized relational database design separating static technical vehicle mappings, real-time physical store inventory, user orders, and administrative staging tables.

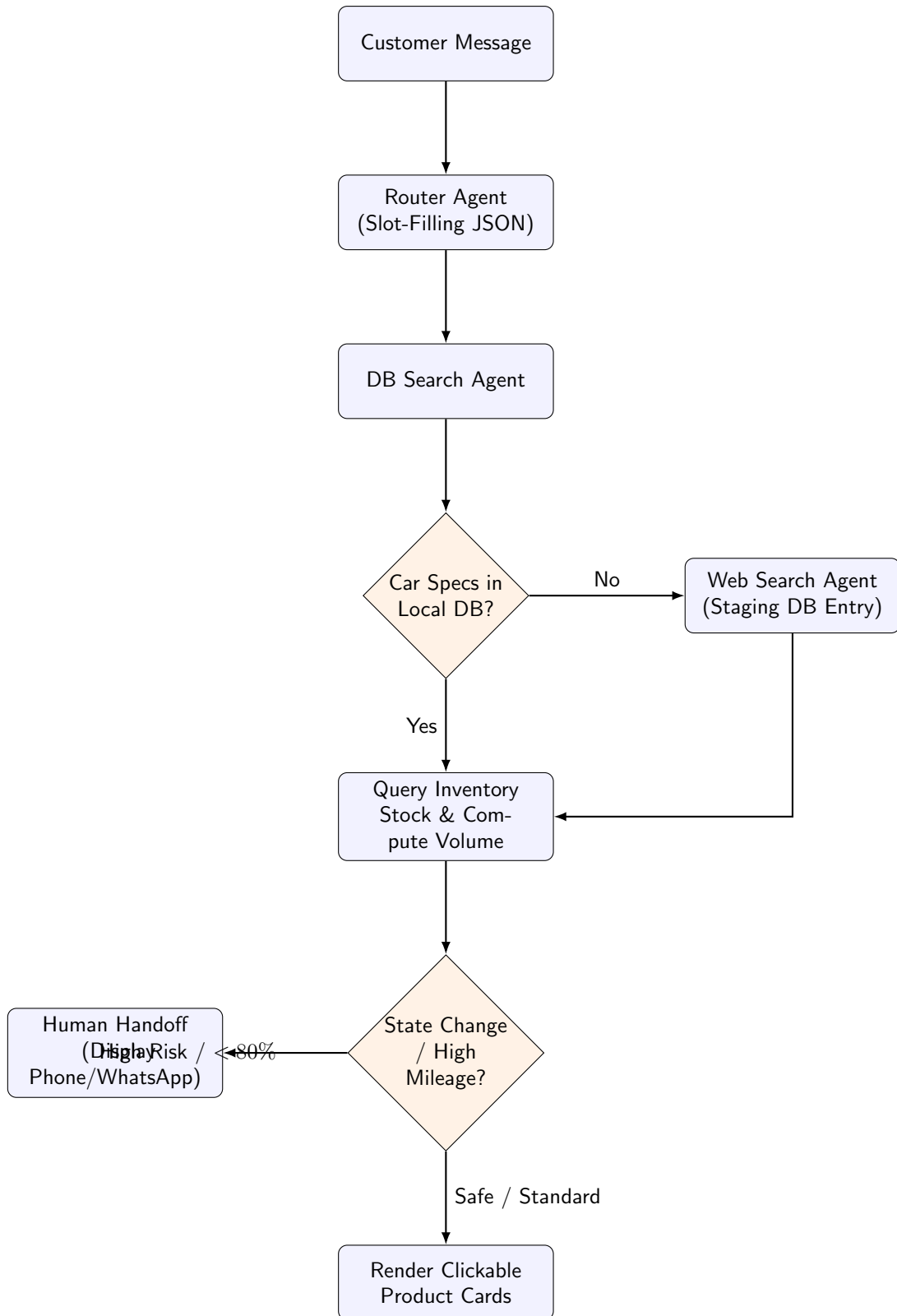


Figure 1: End-to-End Multi-Agent Architecture and Decision Flow.

4.1 Relational Tables

```
-- 1. VEHICLE & TECHNICAL SPECIFICATIONS
CREATE TABLE cars (
```

```

    id INT PRIMARY KEY AUTO_INCREMENT,
    brand VARCHAR(50) NOT NULL,           -- e.g., Renault, Peugeot,
Volkswagen
    model VARCHAR(50) NOT NULL,           -- e.g., Symbol II, 208,
Golf 7
    year_min INT NOT NULL,
    year_max INT NOT NULL,
    engine_reference VARCHAR(50) NOT NULL, -- e.g., 1.2 16V D4F, 2.0
TDI
    fuel_type ENUM('GASOLINE', 'DIESEL', 'LPG') NOT NULL,
    required_viscosity VARCHAR(10) NOT NULL, -- e.g., 5W-40, 0W-30
    oem_specification VARCHAR(50) NOT NULL, -- e.g., RN0700, VW 507.00
    oil_capacity_liters FLOAT NOT NULL,     -- e.g., 4.3
    transmission_type ENUM('MANUAL', 'AUTOMATIC_ATF', 'DSG_DCT', 'CVT')
NOT NULL,
    gearbox_code VARCHAR(50),              -- e.g., BVM5 (JH3), AL4,
DQ200
    trans_oil_spec VARCHAR(50),             -- e.g., 75W-80 API GL-4
    trans_oil_capacity_l FLOAT,
    is_verified BOOLEAN DEFAULT TRUE       -- FALSE for staging records
from web agent
);

-- 2. PHYSICAL STORE INVENTORY
CREATE TABLE inventory (
    id INT PRIMARY KEY AUTO_INCREMENT,
    product_name VARCHAR(150) NOT NULL,
    brand VARCHAR(50) NOT NULL,           -- TotalEnergies, Liqui Moly
, Naftal, Castrol
    category ENUM('ENGINE_OIL', 'TRANS_FLUID', 'BRAKE_FLUID', 'COOLANT'
, 'FILTER') NOT NULL,
    viscosity VARCHAR(10),
    oem_approval VARCHAR(100),
    container_size_liters FLOAT NOT NULL, -- 1.0, 4.0, 5.0
    price_dzd DECIMAL(10, 2) NOT NULL,
    stock_quantity INT NOT NULL DEFAULT 0,
    image_url VARCHAR(255),
    description TEXT
);

-- 3. CUSTOMER ORDERS
CREATE TABLE orders (
    id INT PRIMARY KEY AUTO_INCREMENT,
    customer_name VARCHAR(100) NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    wilaya VARCHAR(50) NOT NULL,
    address TEXT NOT NULL,
    total_price_dzd DECIMAL(10, 2) NOT NULL,
    order_status ENUM('PENDING', 'CONFIRMED', 'SHIPPED', 'CANCELLED')
DEFAULT 'PENDING',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 4. ORDER ITEMS (LINE ITEMS)
CREATE TABLE order_items (
    id INT PRIMARY KEY AUTO_INCREMENT,
    order_id INT NOT NULL,
    product_id INT NOT NULL,

```

```

quantity INT NOT NULL,
unit_price_dzd DECIMAL(10, 2) NOT NULL,
FOREIGN KEY (order_id) REFERENCES orders(id) ON DELETE CASCADE,
FOREIGN KEY (product_id) REFERENCES inventory(id)
);

```

Listing 1: Core Schema Definition for Vehicle Specs, Stock, and Orders

4.2 Container Volume Matching Algorithm

When an engine requires volume V_{req} (e.g., 4.3 L), the DB Search Agent calculates matching inventory options using the following logic:

$$V_{\text{purchased}} = \min \left\{ \sum_i c_i \mid \sum_i c_i \geq V_{\text{req}}, c_i \in S \right\} \quad (2)$$

where $S = \{1.0, 4.0, 5.0\}$ represents the set of available container sizes in liters. The agent prioritizes single-bottle solutions (e.g., 1×5 L) or exact combinations (e.g., 1×4 L + 1×1 L) based on stock availability and total price efficiency in DZD.

5 Conclusion & Next Implementation Steps

This specification provides a comprehensive functional and technical foundation for building a specialized automotive lubricant e-commerce platform customized for Algerian market preferences.

5.1 Immediate Development Milestones

1. **Database Seeding:** Populate the `cars` table with the top 100 most popular Algerian vehicle model/engine combinations using manufacturer technical datasheets.
2. **Agent Prompting & Guardrails:** Implement system prompts for the Router and Evaluator agents, establishing strict JSON schema enforcement for inter-agent communication.
3. **Frontend UI Component Setup:** Build the dual-interface web application (Direct Catalog Filters + Chat Window with dynamic product card rendering and slide-out Shopping Bag).
4. **Admin Dashboard Construction:** Create a lightweight backend interface for store staff to verify web-searched staging entries (`is_verified = FALSE`) and process incoming telephone orders.